

ET5218 INNOVATION AND COMMERCIALIZATION OF TECHNOLOGIES

01 Innovation Management and Entrepreneurship

NTU Innovation and
Entrepreneurship



Course Aims

1. Key issues of success and failure in commercialising inventions/innovations
2. Challenges posed by shortened life cycles of products & services
3. Opportunities posed by emerging technologies
4. Successful transfer from research to commercialisation to global market
5. Values & Skillsets as an entrepreneur/corporate entrepreneur

Course Learning Outcomes

1. Explain the role of innovation in creating a competitive advantage that leads to entrepreneurial wealth creation.
2. Formulate the systematic entrepreneurial activities required by an entrepreneur in New Product Development and New Service Development for sustainable success.
3. Examine the corporate strategy and organizational knowledge of new products and services for a corporate entrepreneur.
4. Appraise the management of intellectual property and strategic alliances in R&D and technology transfer from R&D to the market.
5. Analyze the opportunities, issues and challenges posed by the shortened product and service life cycle and emerging technologies found in the Fourth Industrial Revolution.

Learning Outcomes for Today

1. Discuss the relationship between Science, Innovation and Entrepreneurship.
2. Distinguish the different Types of Innovation and the different Industrial Revolution eras.
3. Appraise the Innovation Management Process.

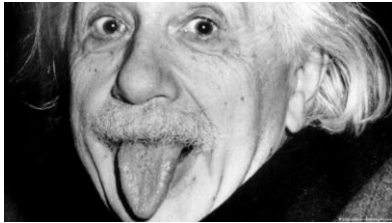
Question

Science $\neq$ Innovation?

Where does Entrepreneurship come in?

Science, Innovation, Entrepreneurship

Albert Einstein



Source: <https://www.dw.com/en/the-story-behind-albert-einsteins-most-iconic-photo/a-56803031> accessed on 7 May 2023

Emmet Lathrop 'Doc' Brown, Ph.D



Source: <https://www.cbr.com/back-to-the-future-doc-brown-best-real-main-character/> accessed on 7 May 2023

Dr. Shi Xu, Nanofilm Technologies International



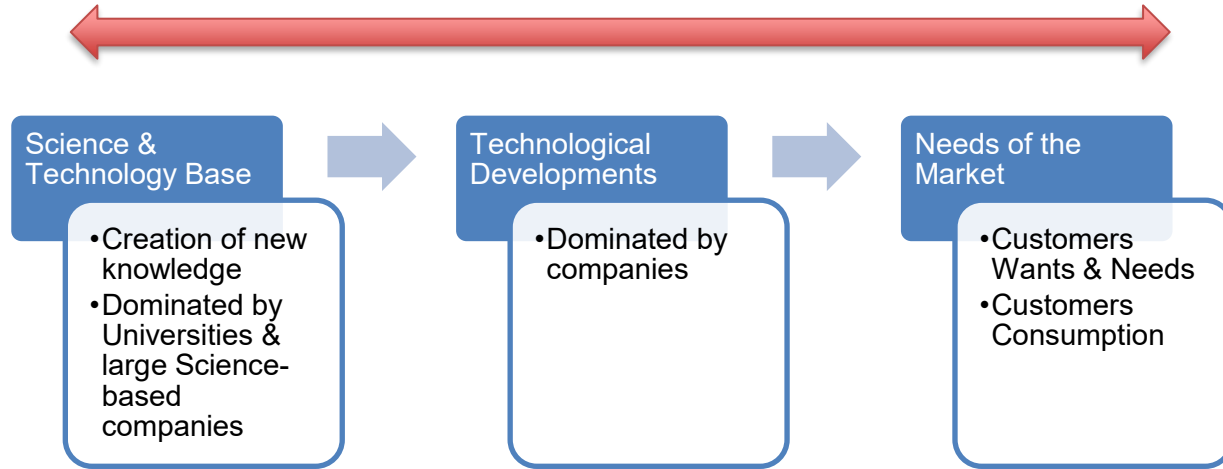
1. The lone inventor/mad professor.
2. Science does not equal innovation;
Innovation is much more than science,
3. Scientific discoveries predate commercial products by many years.
4. Entrepreneurship is more than innovation.

Read:

<https://www.straitstimes.com/business/companies-markets/nanofilm-ipo-turns-former-ntu-professor-into-a-billionaire> accessed on 8 Jun 2023

Conceptual Framework of Innovation

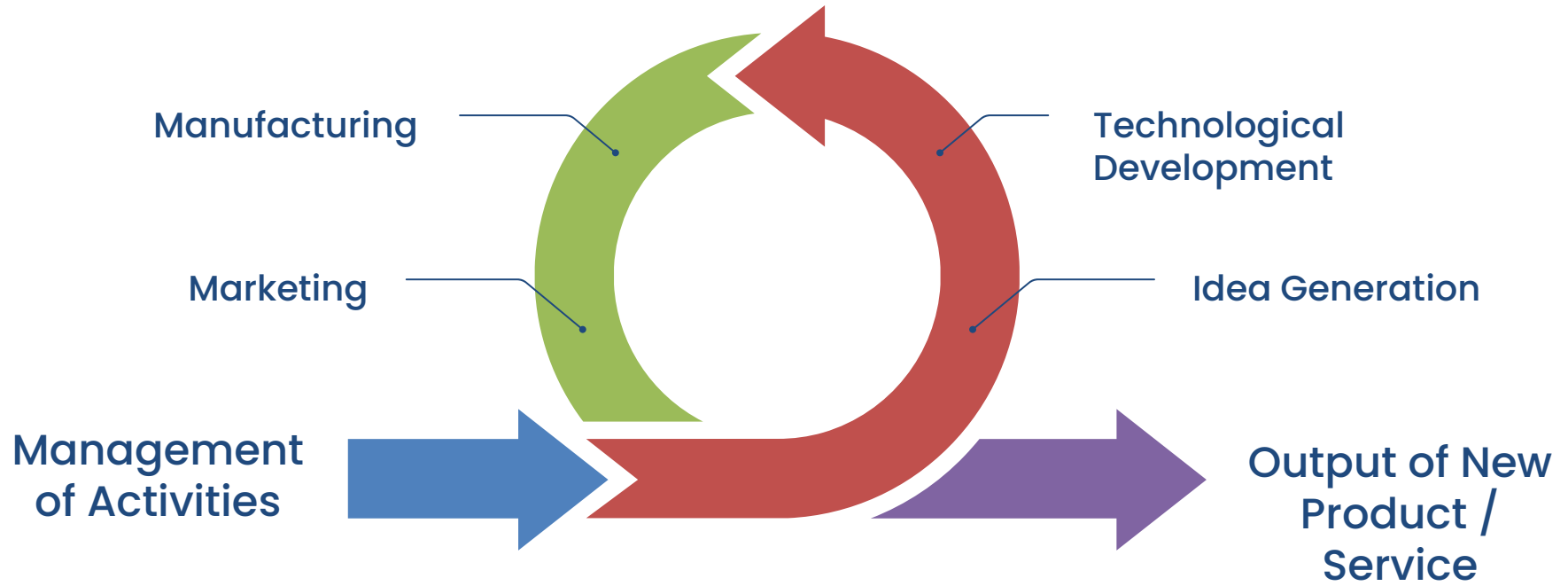
Entrepreneurship comes in to weave the three factors into innovation



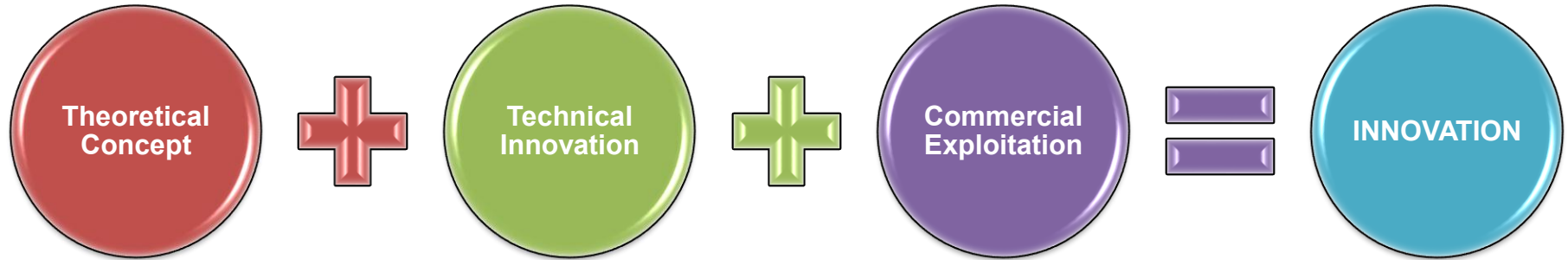
Source:
<https://www.forbes.com/sites/bernardmarr/2024/11/11/4-smartphone-with-the-best-artificial-intelligence/> accessed 8 Aug 2025

Source: Conceptual Framework of Innovation [Adapted from Trott, P. (2021)]

Innovation Management



Innovation Equation



Source: Innovation Equation [Adapted from Henry and Walker (1991)]

“...its engineers had developed a "fantastic electric car" but that it would not hit the roads because it was not "commercially viable".



Source: <https://www.bbc.com/news/business-50004184>
accessed on 11 Oct 2019

Types of Innovation

Let's understand the different types of innovation that an entrepreneur / corporate entrepreneur can be involved in

Product Innovation



Source: <https://sxfi.com/products/>
accessed on 6 Jan 2022

Process Innovation



Source: <https://www.3ders.org/articles/20170216-nasa-launches-first-ever-3d-printed-space-art-into-orbit.html>
accessed on 6 Jan 2022

Types of Innovation

Marketing Innovation

- Leasing of Photocopier
- Pre-paid SIM Card

The Ultimate Expat Guide to
GETTING THE BEST PREPAID SIM CARD

INTERNATIONAL CALLS	STAYING LESS THAN 3 MONTHS	STAYING MORE THAN 3 MONTHS
China	\$8 hot SIM \$8 Prepaid SIM	\$15 Happy Prepaid SIM
Bangladesh Myanmar Philippines	LOCAL DATA	LOCAL CALLS
India Indonesia	For <1GB/mth For >1GB/mth	22 cents for first minute 8 cents per minute

Read on for full details...

Source: <https://blog.seedly.sg/expat-guide-prepaid-sim-card-singapore/> accessed on 6 Jan 2022

Service Innovation

GrabFood
Crave it? Grab it.
Want to be a GrabFood Merchant-partner?
Click [here](#).



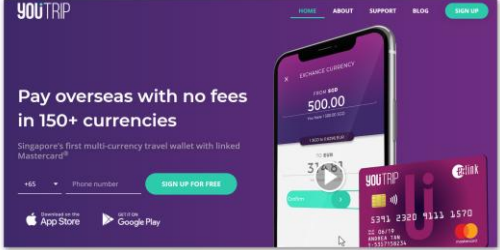
Your favourite food delivered hot and fresh

Source:
<https://www.grab.com/sg/food/>
accessed on 6 Jan 2022

YOU TRIP

Pay overseas with no fees
in 150+ currencies

Singapore's first multi-currency travel wallet with linked Mastercard®



Download on the App Store GET IT ON Google Play

Sign up for free

Source:
<https://www.you.co/en-SG/>
accessed on 6 Jan 2022

Types of Innovation

Organisational Innovation



Source:

<https://www.channelnewsasia.com/cna-insider/how-fujifilm-survived-digital-age-unexpected-makeover-1026656>

on 6 Jan 2022



Source:

<https://www.thecanadianencyclopedia.ca/en/article/blackberry-limited>

accessed on 6 Jan 2022



Source: Retrieved from

https://www.singaporeair.com/en_UK/us/flying-withus/our-story/our-subsidiaries/

accessed on 12 Sep 2021

Types of Innovation

Management Innovation

- 5S (Sort, Set in Order, Shine, Standardise, Sustain)
- 6 Sigma (or Zero Defects)
- Total Quality Management

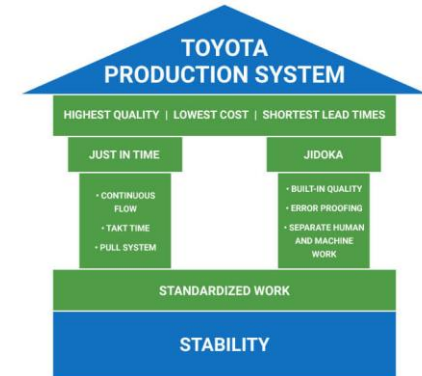
5S Summary

Seiri	Seiton	Seison	Seiketsu	Shitsuke
				
Sort	Set in order	Shine	Standardize	Sustain
Remove all of the clutter from the work place	Organize in an efficient and ergonomic manner.	Clean up the entire area removing all dirt	Ensure standard ways of working for the first three stages.	Ensure that 5S principles are part of the culture

Source: <https://www.qualitygurus.com/5s-for-workplace-improvement/>
accessed on 6 Jan 2022

Production Innovation

- Quality Circles
- Just-In-Time
- Jidoka



Source: <https://blog.5stoday.com/the-toyota-production-system-house/> accessed on 6 Jan 2022

Industry 1.0 (1760 -1840)

Innovation	Innovator	Date
Steam Engine	James Watt	1770-80
Iron Boat	Isambard Kingdom Brunel	1820-45
Locomotive	George Stephenson	1829
Electromagnetic Induction Dynamo	Michael Faraday	1830-40

Industry 2.0 (1870 –1914)

Industry Innovative new products or services

Electricity Light bulb
Factories
Telegram (Morse Code)

Transport Railroad

Industry 3.0 (1945 – 2011?)

Industry	Market leaders	Innovative new products and services
Computers and software development	Intel; IBM; Lenovo; Microsoft; SAP	Computer chip technology, computer hardware improvements and software development
Motorcars	Toyota; BMW; Hyundai	Car design and associated product developments
Smart phones	Samsung; Apple; Huawei; Oppo; Xiaomi	Design and new features
Internet-related industries	Yahoo; Google; Ebay	New ecommerce services
Pharmaceuticals	Pfizer; GlaxoSmithKline; Moderna	Blood pressure; Contraception; Impotence; ulcer treatment drug

Industry 4.0 (? 2011 or the Future)

- **4th Industrial Revolution Themes**
- Interconnection between man & machines through IoT and IoP (Internet of People)
- Comprehensive information available for data analytics
- Technological Assistance to aid human decision-making, problem solving & perform unsafe tasks
- Decentralised decisions by cyber physical systems for autonomous tasks

Sources:

1. Avila, O. (2020). Industry 4.0: The fourth industrial revolution. <https://www.linkedin.com/pulse/industry-40-fourth-industrial-revolution-oscar-avila/> accessed 7 Jun 2023.

The Future Digital Paradigm

	Time	1994	2004	2014	Future
	Analog World	Internet/Web	E-Business Digital Marketing	Digital business Internet of Things	Semi-autonomous machines Robots
Focus	Relationship with humans for a better experience	Expanded relationship in new markets & countries	Transformation of customer interaction into a global and efficient medium	Expansion of the relationship of people to machines	Intelligent, (semi-) autonomous machines interact with people and social systems
Design Criteria	• Needs • Simple • Functionality	• Networking • Availability • Data	• Information • Business Intelligence • Big Data	• Knowledge • Prediction • Access to sensors	• Trust • Adaptability • Intention
Systems	• People	• People • Web	• People • Cloud	• People • Sensors • Objects	• People • Machines • Robots • Social systems • Culture
Results	Optimised relationship	Expanded Relationship	Optimised channels and interactions	New business models	Close relationship, Human-machine as a team

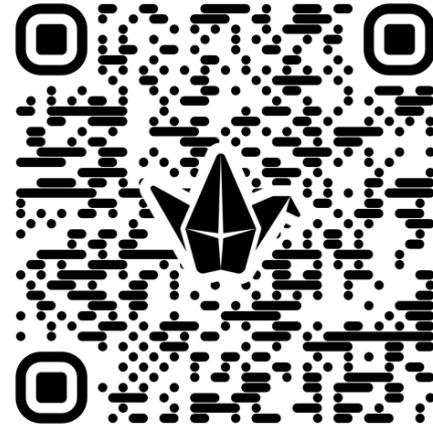
Source: Lewrick, M., Link, P., Leifer, L.. (2018). *The Design Thinking Playbook*. Wiley, p 267.

Discussion 1.1

Questions

1. What other future invention/innovation could potentially drive economic development in the future or at Industry 4.0?

2. And how will it happen?



Klein Vision AirCar

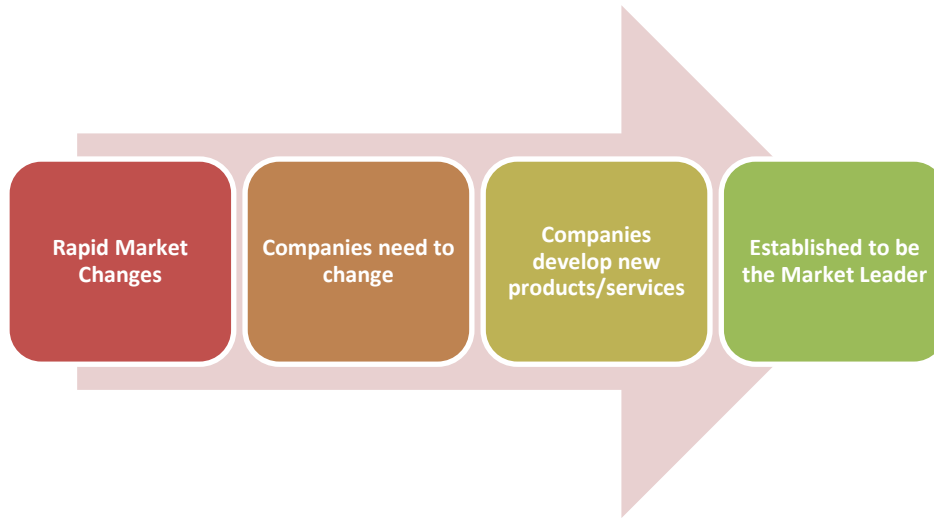


Joby Aviation x Toyota

Source: <https://www.topgear.com/car-news/best-2022/will-one-these-flying-cars-finally-change-game-2023> accessed on 8 May 2023

A Flying Car?

Importance of Innovation



ByteDance
字节跳动

Grab NETS

Meta



vs

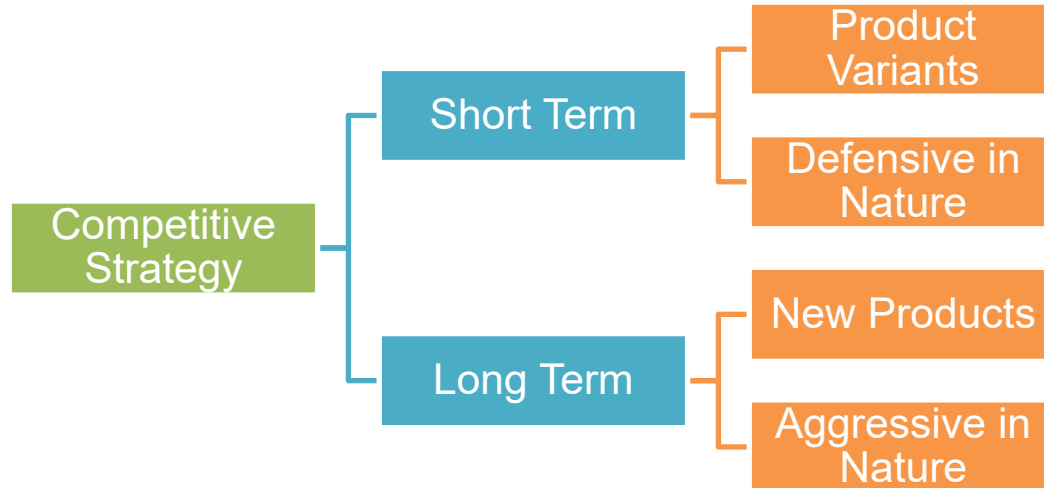


AIRBUS

BOEING

COMAC

Product / Service Strategy



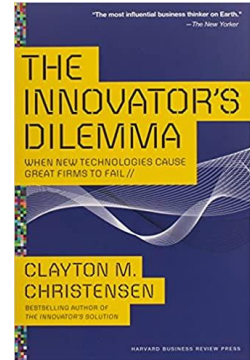
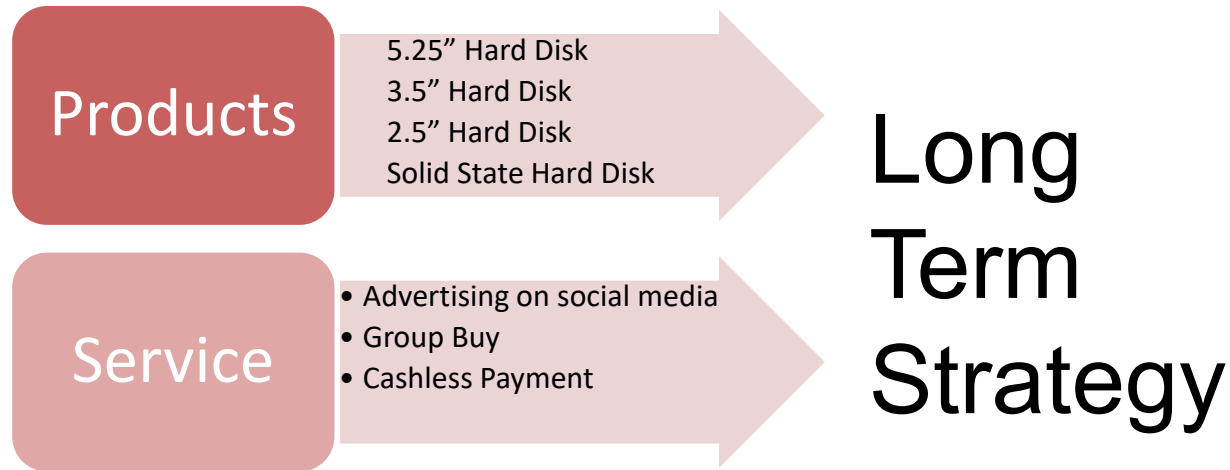
Sustaining Innovation

Disruptive Innovation



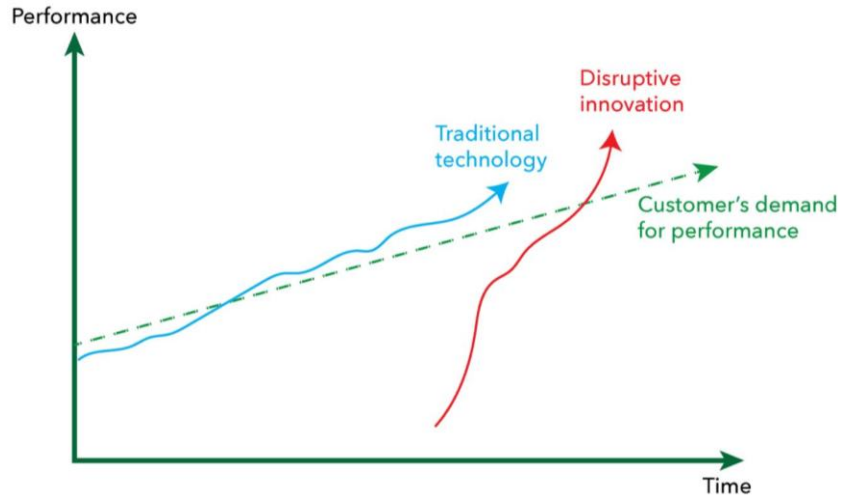
A term coined by Prof Clayton Christensen (2003)

Radical innovations that create new markets and significant changes especially for consumers



Source:
<https://www.amazon.sg/>

Disruptive Innovation



Source: Disruptive Innovation [Adapted from Trott, P. (2021)]



Source:

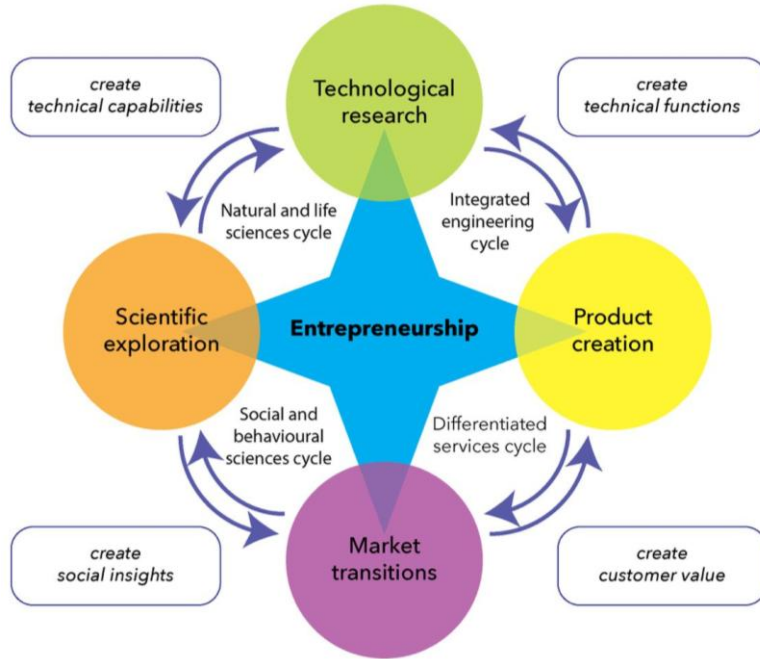
<https://static.mothership.sg/1/2023/05/punggol-barista-spill.png> accessed on 8 May 2023



The autonomous robot moves around the loo seeking out stains and areas to clean, guided by sensors. © PHOTO: HO JOO SUE

Source: <https://www.straittimes.com/tech/autonomous-cleaning-bot-to-start-scrubbing-public-toilets-in-early-2024> accessed on 12 Dec 2023.

Cyclic Innovation Model



- The innovation circle with interconnected cycles.
- Behavioural sciences and engineering as well as natural sciences and markets are brought together.
- Central to this model is Entrepreneurship. **Without entrepreneurship, there is no innovation**, and **without innovation there is no new business.**

Source: Cyclic Innovation Model. [Adapted from: Berkhout, A.J., Hartmann, D. and Trott, P. (2010)]



Example Cyclic Innovation Cycle

<https://www.reuters.com/business/healthcare-pharmaceuticals/kariko-thc-care-pharmaceuticals/kariko-weissman-win-medicine-nobel-covid-19-vaccine-work-2023-10-02/>
accessed 6 Oct 2023

- Central to this model is **Entrepreneurship**.
- Without entrepreneurship, there is no innovation**, and **without innovation there is no new business**.



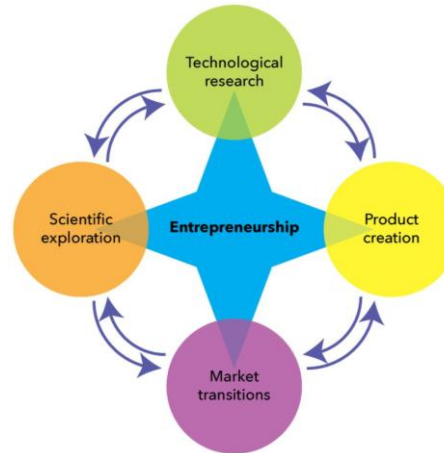
Dr. Drew Weissman

Source: <https://mainnovation.med.upenn.edu/people/drew-weissman-m-d-ph-d/> accessed 7 Aug 2024



Dr. Kati Kariko

Source: Retrieved from <https://www.nytimes.com/2021/04/08/health/coronavirus-mrna-kariko.html> on 29 Aug 2021



Source: Cyclic Innovation Model. [Adapted from: Berkhout, A.J., Hartmann, D. and Trott, P. (2010)]

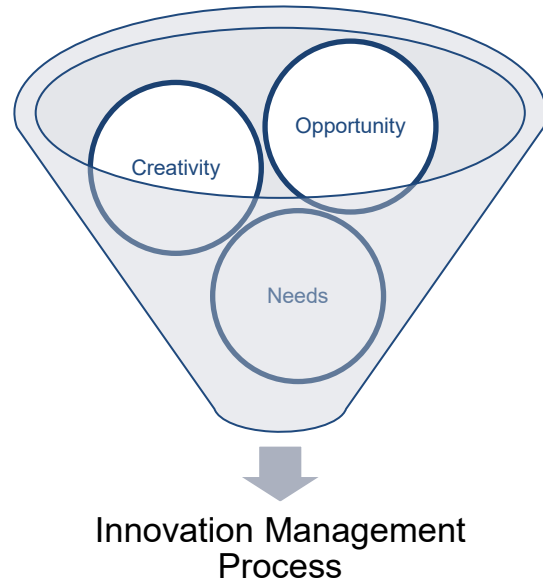
Source: Retrieved from <https://www.designweek.co.uk/issues/4-10-january-2021/pfizer-rebrand/> on 14 Aug 2024



Dr. Ugur Sahin and Dr. Özlem Türeci

Source: Retrieved from <https://time.com/5927342/mrna-covid-vaccine/> on 1 Mar 2020

Innovation Management Process

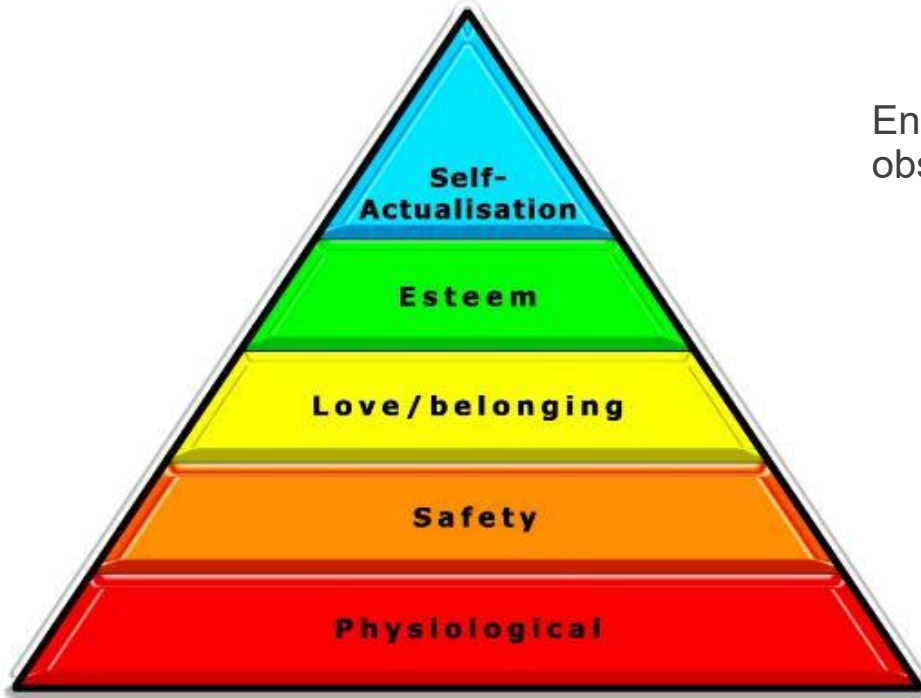


Entrepreneurship /
Corporate Entrepreneurship
/ Social Entrepreneurship

Maslow Hierarchy of Needs



Entrepreneurial Opportunities may arise
observing society's Maslow Hierarchy of Needs



Maslow, A.H. (1943). "A theory of human motivation". *Psychological Review*. 50 (4): 370–96.

Needs – Physiological



Air: Need to Breathe
(Xiaomi Air Purifier)



Source: www.mi.com

Water: Need for Drink
(Philips On Tap Water Purifier)



Source: www.philips.com.sg

Food: Need for
Nourishment
(Xndo Low Carb, High
Protein & Trans-fat Free)



Source: Xndo.com

Needs - Safety



Personal Safety:
Personal Safety Devices



Source: Alibaba.com

Finance Safety:
Insurance



Source: various

Health & Well-Being: Accessories
(3M N95 Masks Phiten Titanium Tape)



Source: various

Needs - Love and Belonging



Friendship & Family:

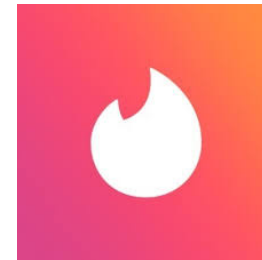
Help to organize picnics for busy families



Source: Straits Times. (2016). Retrieved from Straits Times on 12 Feb 2016

Intimacy:

Tinder, Coffee Meets Bagel, Paktor, OK Cupid



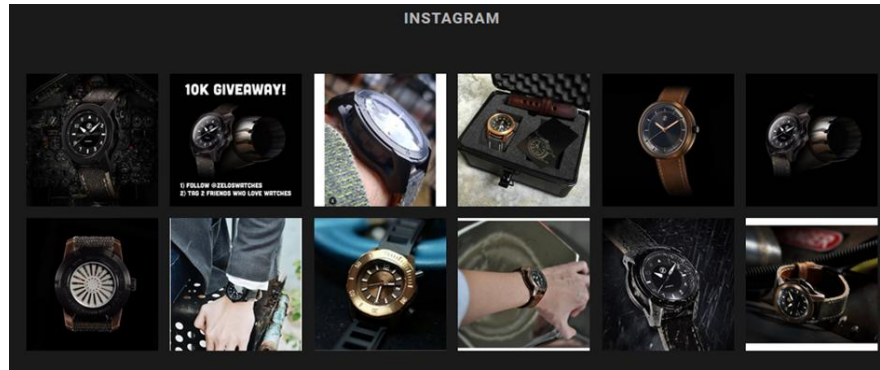
Source: various

Needs - Esteem



Human Desire: Luxury Watches Market

Source: Zelos Watches. (2016). Retrieved from <http://zeloswatches.com/> on 28 Jan 2016



About Us

Zelos was founded to offer timepieces with cutting-edge designs and materials. We design and produce watches from the ground up, ensuring that you get a watch that can be found nowhere else.

Since the start of Zelos in 2014, we have launched 3 models on Kickstarter with increasing success. Our first model, the Helmsman garnered 273% of the targeted funding. The Chroma followed next with 350% and finally the Abyss with an overwhelming 650% of funding.

Zelos is based in Singapore and was founded by Elshan. Having been fascinated by mechanical timepieces for many years, building one was the next logical step. He believes that great timepiece designs should not be confined to luxury brands.



Needs – Self Actualisation



Altruism: Recycling and Reuse Plastic in Concrete



Source: Straits Times. (2016). Retrieved on 11 Jan 2016.

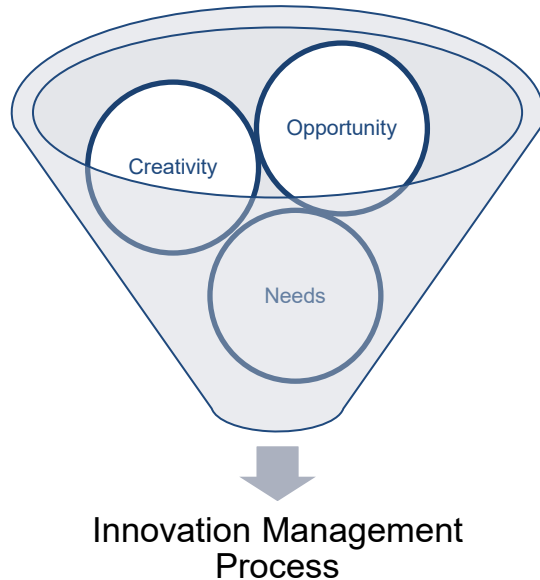
Altruism: Humanity Poor Indonesians swap rubbish for healthcare

Dr Gamal Albinsaid came up with idea of a system where people bring their waste to clinics as payment for health care service. He did this after hearing how the daughter of a rubbish scavenger died from diarrhoea having no access to the most basic healthcare.



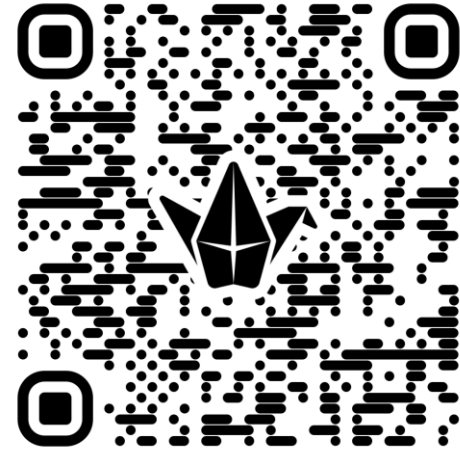
Source: BBC News. (2018). Retrieved from <http://www.bbc.com/news/av/world-asia-26227145/poor-indonesians-swap-rubbish-for-healthcare> accessed 7 Jan 2018

Discussion 1.2



Questions:

1. What are some of society's current/future needs?
2. What are some of the current/future opportunities?
3. What are some of the creative ways of delivery the products or services?



Recommended Readings

- Lewrick, M., Link, P., Leifer, L.. (2018). *The Design Thinking Playbook*. Wiley: 266 - 267
- Maslow, A.H.. (1943). A theory of human motivation. *Psychological Review*. 50 (4): 370–396.

End